ZeroUnlearn: Few-Shot Knowledge Unlearning in Large Language Models

Supplementary Experimental Results

A Evaluation on RWKU

Table 1: Unlearning results on RWKU (Level 1). NPO fine-tunes three layers, consistent with our method, whereas NPO-full fine-tunes all layers.

Method	**Eff.↓**	**PPL↓**
Base	39.66±0.17	12.87
GA	**1.62**±17.1	976.00±6.3
FT	41.35±0.95	13.44±0.14
ROME	38.72±2.83	12.90±0.08
MEMIT	37.99±1.22	12.86±0.02
AlphaEdit	37.17±2.12	12.81±0.04
NPO	39.07±0.59	12.93±0.01
NPO-full	35.91±1.48	13.22±0.14
ZeroUnlearn	35.70±3.79	13.18±0.18

Table 2: Unlearning results on RWKU (Level 2).

Method	**Eff.↓**	**PPL↓**
Base	39.12±0.17	12.87
GA	**1.56**±2.8	>1000
FT	38.09±1.02	12.92±0.07
ROME	38.51±1.77	12.91±0.09
MEMIT	38.01±1.42	12.87±1.80
AlphaEdit	35.53±2.82	12.86±0.06
NPO	38.98±0.27	12.89±0.01
NPO-full	35.56±2.51	13.00±0.09
ZeroUnlearn	35.47±2.40	13.10±0.14

Table 3: Unlearning results on RWKU (Level 3).

Method	**Eff.↓**	**PPL↓**
Base	39.12±0.17	12.87
GA	**2.64**±4.53	> 1000
FT	40.62±1.56	13.01±0.06
ROME	39.07±2.88	12.92±0.09
MEMIT	38.93±1.99	12.86±1.30
AlphaEdit	37.45±2.62	12.84±4.10
NPO	39.71±0.43	12.91±0.01
NPO-full	37.66±2.36	13.05±0.08
ZeroUnlearn	34.01±4.75	13.10±0.18

B Performance of NPO on Knowledge Editing Datasets

Table 4: Unlearning results of NPO on the MCF dataset. NPO fine-tunes three layers, consistent with our method, whereas NPO-full fine-tunes all layers.

Method	**Eff.↓**	**Gen.↓**	**Spe.↑**	**PPL↓**
NPO	17.00±4.40	18.30±4.73	17.90±3.30	12.92±0.12
NPO-full	16.44±4.69	18.22±5.88	**18.40**±4.05	13.03±0.11
ZeroUnlearn	**0.40**±0.80	**4.60**±2.24	14.90±2.93	13.06±0.18

Table 5: Unlearning results of NPO on the ZsRE dataset

Method	**Eff.↓**	**Gen.↓**	**Spe.↑**	**PPL↓**
NPO	30.77±3.72	30.15±4.24	**28.09**±2.47	12.91±0.96
NPO-full	30.44±4.09	29.90±3.88	27.71±2.67	13.03±0.07
ZeroUnlearn	**27.85**±3.87	**27.52**±3.87	27.73±2.70	13.08±0.06

Table 6: Unlearning results of NPO on the MQUAKE dataset

Method	**Eff.↓**	**PPL↓**
NPO	32.33±5.23	12.9±0.01
NPO-full	30.64±4.63	12.93±0.07
ZeroUnlearn	**24.51**±3.37	13.15±0.16

C Sample Size Sensitivity of ZeroUnlearn

Table 7: Unlearning results on the `MCF` dataset under different sample sizes.

# Forgotten Samples	**Eff.↓**	**Gen.↓**	**Spe.↑**	**PPL↓**
10	1.00±3.00	7.00±5.09	20.60±4.86	13.10±0.11
25	0.80±1.60	5.00±2.86	17.52±3.71	13.06±0.13
50	0.40±0.80	4.60±2.24	14.90±2.93	13.06±0.18
100	0.50±0.81	5.15±1.78	14.72±2.31	13.05±0.19
200	0.90±0.70	5.90±0.77	13.29±0.86	13.25±0.15
500	3.25±0.43	6.22±0.25	12.53±0.78	13.77±0.38
1000	6.73±0.42	8.45±0.66	12.20±0.49	14.25±0.41

Table 8: Unlearning results on the `ZsRE` dataset under different sample sizes.

# Forgotten Samples	**Eff.↓**	**Gen.↓**	**Spe.↑**	**PPL↓**
10	24.82±13.60	25.67±13.69	28.51±7.47	13.07±0.07
25	24.97±5.39	25.60±5.92	27.25±4.79	13.13±0.84
50	27.85±3.87	27.52±3.87	27.73±2.70	13.08±0.06
100	26.74±2.13	27.17±1.77	27.78±2.73	13.15±0.10
200	26.53±1.82	26.54±2.01	26.98±1.73	13.18±0.12
500	24.23±0.11	23.76±0.28	24.39±1.01	13.69±0.35
1000	20.67±0.34	19.85±0.25	20.48±0.44	14.50±0.55

Table 9: Unlearning results on the `ZsRE` dataset under different sample sizes.

# Forgotten Samples	**Eff.↓**	**PPL↓**
10	24.98±9.91	13.05±0.06
25	24.67±6.72	13.03±0.09
50	24.51±3.37	13.15±0.16
100	23.01±2.53	13.18±0.20
200	21.75±2.28	13.27±0.27
500	21.56±1.66	13.31±0.19
1000	19.28±1.34	13.61±0.22

D Ablation Study of Orthogonal Projection and Utility Terms

Table 10: Ablation results of on the ZsRE dataset.

Method	Eff.↓	Gen.↓	Spe.↑	PPL↓
ZeroUnlearn	27.85±3.87	27.52±3.87	27.73±2.70	13.08±0.06
w/o OP	28.45±4.26	29.08±4.72	27.99±2.69	12.98±0.09
w/o UT	23.65±3.14	24.89±3.25	25.28±2.33	14.98±0.35

Table 11: Ablation results of on the MQUAKE dataset.

Method	Eff.↓	PPL↓
ZeroUnlearn	24.51±3.37	13.15±0.16
w/o OP	27.83±4.11	12.87±0.07
w/o UT	21.97±3.17	15.14±0.27

E Further Exploration of Neutral Target $\mathbf{M}_n$

Table 12: Results on the MCF dataset under different $\mathbf{M}_n$

$\mathbf{M}_n$	Eff.↓	Gen.↓	Spe.↑	PPL↓
"<EOS>"	0.40±0.80	4.60±2.24	14.90±2.93	13.06±0.18
"I don't know."	1.00±1.34	5.70±1.67	16.38±3.50	13.15±0.96
"Hello."	0.20±0.60	4.80±2.36	14.90±2.70	13.21±0.12

Table 13: Results on the ZsRE dataset under different $\mathbf{M}_n$

$\mathbf{M}_n$	Eff.↓	Gen.↓	Spe.↑	PPL↓
"<EOS>"	27.85±3.87	27.52±3.87	27.73±2.70	13.08±0.06
"I don't know."	27.01±2.91	26.81±3.44	27.24±2.70	13.06±0.12
"Hello."	27.14±2.75	27.50±3.12	26.99±2.34	13.09±0.15

Table 14: Results on the MQUAKE dataset under different $\mathbf{M}_n$

$\mathbf{M}_n$	Eff.↓	PPL↓
"<EOS>"	23.60±3.07	13.11±0.17
"I don't know."	22.66±3.04	13.14±0.10
"Hello."	24.51±3.37	13.15±0.16

F Computational Cost Across Sample Sizes

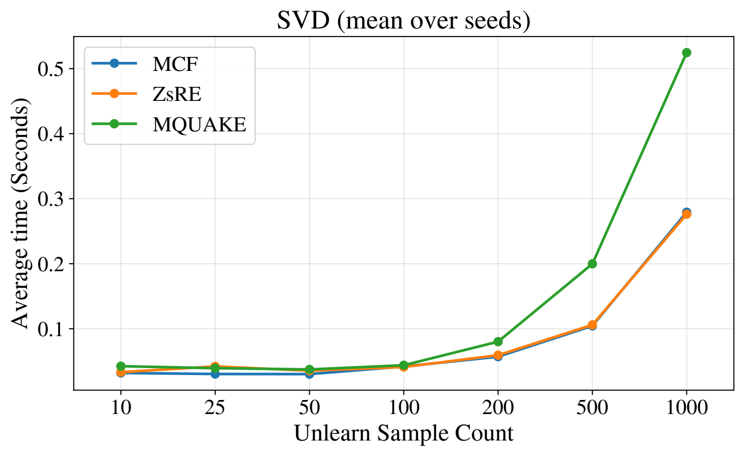

Figure 1: Computational cost across sample sizes for SVD component.

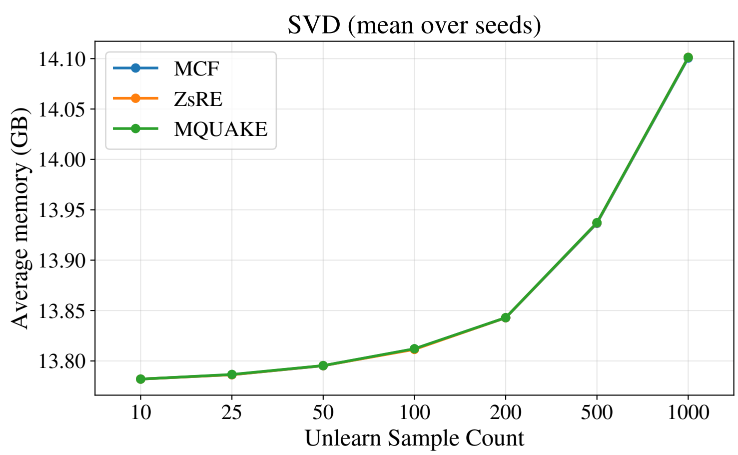

Figure 2: Peak memory consumption across sample sizes for SVD component.

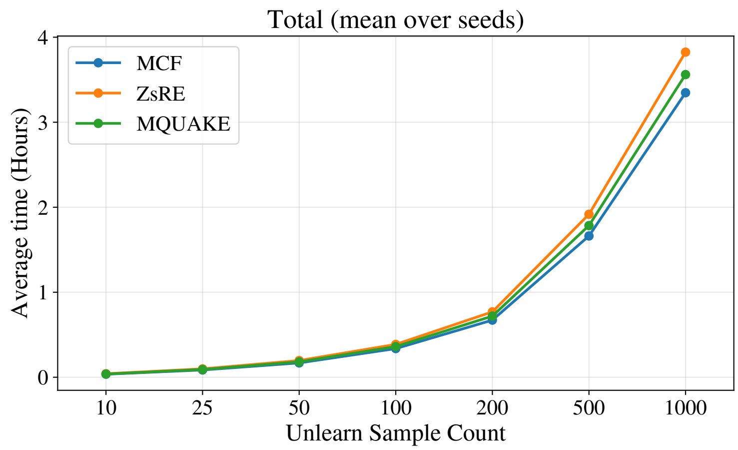

Figure 3: Computational cost across sample sizes for ZeroUnlearn.

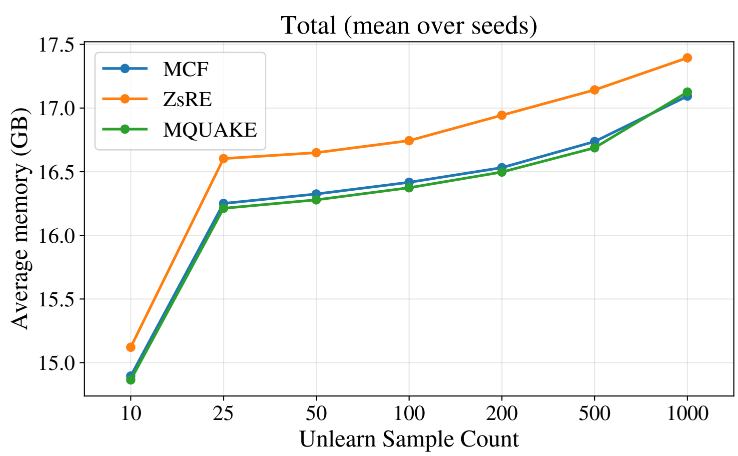

Figure 4: Peak memory consumption across sample sizes for ZeroUnlearn